VEERA MALLA REDDY. V

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AI/ML - Data Project Management Leader | PMP®

Technically adept architect and engineer with 18+ years of experience designing and implementing scalable data systems, with the last 4 years **deeply focused on productionizing Generative AI and Agentic AI solutions**. Proven expertise in architecting **RAG (Retrieval-Augmented Generation) pipelines**, building and optimizing **vector databases (Pinecone, FAISS)** for real-time AI inference, and implementing **Model Context Protocols (MCPs)** for seamless LLM-backend integration. Adept at collaborating with AI/ML and backend teams to deliver contextual, compliant, and high-performance AI systems on **cloud-native platforms (AWS, Azure)**. Strong background in data governance and performance tuning within global, regulated financial environments.

CORE COMPETENCIES & TECHNICAL SKILLS

Area	Technologies & Expertise
Generative AI & LLMs	RAG Architecture, Agentic AI, Prompt Engineering, Model Context Protocols (MCPs), LLM Evaluation, Fine-tuning strategies
AI Frameworks & Tools	LangChain, LlamaIndex, Semantic Kernel, OpenAI API (GPT-4, GPT-4-turbo), Anthropic Claude API, Hugging Face Transformers
Vector Databases	Pinecone, Qdrant, Milvus, FAISS, ChromaDB – Indexing, Embedding, Similarity Search, Hybrid Search
ML/DL Frameworks	TensorFlow, PyTorch, Scikit-learn, MLflow
Cloud AI Services	AWS (SageMaker, Bedrock) , Azure (OpenAI Service, ML Studio, Cosmos DB) , GCP (Vertex AI)
Core Databases	PostgreSQL (Performance Tuning, Extensions) , MongoDB, Cassandra, Redis, Oracle, Greenplum
Data Engineering & Big Data	Spark, Kafka, Hadoop Ecosystem (Hive, HDFS) , ETL/ELT Pipelines, Data Governance, Data Lineage, Real-time Processing
Programming & APIs	Python (Pandas, NumPy, FastAPI, SQLAlchemy), Scala, SQL, Shell Scripting
Cloud & DevOps	AWS (S3, Glue, Lambda, Kinesis, RDS, Aurora), Azure (Data Factory, ADLS, Synapse), CI/CD (Azure DevOps, GitLab) , IaC (Terraform, Ansible), Docker, Kubernetes
Compliance & Methodologies	GDPR, SOC 2, BCBS 239, Data Retention Policies, Agile (Scrum), SAFe, DevSecOps

GENERATIVE AI & STRATEGIC PROJECTS

- **Generative AI-Powered Knowledge Assistant (Standard Chartered - POC Lead):** Architected and led the development of a RAG-based Q&A system over 10,000+ regulatory documents. **Tech Stack:** LangChain, Azure OpenAI (GPT-4), Pinecone, Azure ADLS, FastAPI. **Impact:** 70% reduction in compliance research time; became a blueprint for the bank's enterprise AI strategy.

- **Agentic AI Framework for Customer Service (Standard Chartered - Technical Architect):** Designed and implemented a proof-of-concept for an AI agent using Model Context Protocols (MCPs). The agent could interpret customer queries, securely retrieve context from core banking systems (PostgreSQL), and suggest next-best actions. **Impact:** Demonstrated a path towards hyper-personalized, automated customer service, influencing the bank's investment in agentic AI.

- **Real-Time Embedding Pipeline for Feature Store (Uber - Technical Lead):** Led the technical design for a proof-of-concept pipeline to generate and serve real-time embeddings for user and trip attributes using Spark Structured Streaming, integrating them into Uber's online feature store. **Impact:** Enabled low-latency ML model inference for real-time personalization features.

PROFESSIONAL CERTIFICATIONS

- PMP®, Project Management Institute
- Data Landscape of GenAI for Project Managers from PMI®
- Practical Application of Generative AI for Project Managers from PMI®
- Cloudera Certified Developer for Apache Hadoop (CCDH)Apache Hadoop (CCDH)

PROFESSIONAL EXPERIENCE

Ingestion Framework Lead – Assistant Vice President, Standard Chartered Bank – GBS, Chennai, India (Apr 2021 – Present)

- **Technical Leadership & Strategy:** Serve as the lead architect and subject matter expert for the bank's AI data infrastructure, providing technical direction to a team of 8-10 data engineers. Define architectural best practices, conduct code reviews, and mentor engineers on RAG pipelines, vector databases, and MCP implementation, elevating the team's overall AI engineering capability.
- **Architected AI-Ready Data Foundation:** Designed and led the implementation of an enterprise ingestion framework processing >5TB of data daily, specifically structured to serve as the **"single source of truth" for all AI/ML and Generative AI initiatives**. The architecture seamlessly integrates batch (ADF, Spark) and real-time (Kafka, IBM CDC) data into a governed Delta Lake on Azure, optimized for both model training and inference.
- **Generative AI RAG Pipeline Architecture (Strategic POC):** Spearheaded a high-impact strategic POC to build a **Regulatory Knowledge Assistant**. Architected a full **RAG pipeline** using **LangChain** to index and retrieve from 10,000+ complex financial regulatory documents stored in ADLS. Integrated **Azure OpenAI Service (GPT-4)** for natural language Q&A and **Pinecone** as the vector database for semantic search. **The POC reduced compliance research time by over 70%** and directly shaped the bank's enterprise AI roadmap.
- **Pioneered Agentic AI & MCP Integration:** Collaborated with AI product teams to design and implement **Model Context Protocols (MCPs)** , creating a reusable framework for secure, contextual communication between LLMs and backend core banking systems. This enabled the development of a proof-of-concept **"Agentic AI" for customer service** that could reason over customer data and suggest next-best actions.
- **Engineered Context Store & Embedding Repository:** Designed and built scalable **context stores** on **PostgreSQL** and **embedding repositories** on **MongoDB and Pinecone** to serve real-time inference requests for GenAI automation modules, improving retrieval latency by over 40% for downstream applications.
- **Established Data Governance for AI Compliance:** Designed and enforced data lineage, retention, and governance policies for all data feeding AI models, ensuring full compliance with **GDPR, SOC 2, and BCBS 239**. Implemented automated data quality checks using Great Expectations, increasing data trust for data scientists by over 60%.

Project Manager – Big Data, Uber India Research and Development, Bengaluru, India (Dec 2020 – Apr 2021)

- **Led Technical Delivery for ML Feature Platform:** Managed the end-to-end technical delivery of critical big data projects that powered Uber's machine learning models (pricing, ETA, personalization). Oversaw the development and optimization of petabyte-scale Spark pipelines that generated core feature tables for the company's **centralized feature store**.
- **Performance Engineering for AI Workloads:** Spearheaded a performance tuning initiative across the Spark/Hadoop platform, specifically targeting ML training data pipelines. Optimized complex PySpark and Scala jobs, **reducing average runtime by 35% and compute costs by 20%** , which accelerated model experimentation cycles for data science teams.
- **Drove Data Lineage Initiative for Model Governance:** Led the project to integrate data lineage and metadata management tools into the data platform, creating transparency and traceability for all data used in model training and inference. This was critical for model governance, debugging, and compliance.
- **Executive Technical Reporting:** Synthesized complex technical analyses of platform health and project outcomes into clear, data-driven reports for engineering leadership and business stakeholders, directly influencing strategic investment decisions for the ML platform.

Lead Consultant - Genpact, Hyderabad, India (Sep 2015 – Dec 2020)

- **Project Leadership & Strategy:** Led the end-to-end project management of large-scale data modernization initiatives, including the design and deployment of scalable Hadoop data pipelines, which enhanced operational efficiency and advanced analytics capabilities for processing high-volume machine data.
- **Cloud Migration Program Management:** Spearheaded the cloud transformation program, successfully migrating enterprise applications and databases to AWS and hybrid platforms by establishing project plans, managing budgets and timelines, and ensuring secure, compliant execution.

- **Cross-Functional Team Coordination:** Directed cross-functional teams comprising development, operations, and infrastructure specialists to automate deployment processes, resulting in reduced project delivery times and enhanced system reliability.
- **Stakeholder & Vendor Management:** Collaborated with business stakeholders and cloud vendors (AWS) to define project requirements, manage expectations, and deliver cloud-native data solutions that aligned with long-term strategic goals.
- **Operational Excellence & Support:** Managed the project for end-to-end data platform support, overseeing the installation, configuration, and maintenance across environments, which minimized downtime and ensured high availability for critical business operations.

Project Leader - L&T Technology Services Limited, Bengaluru, India (Apr 2014 – Sep 2015)

- **Infrastructure Project Delivery:** Project-managed the architecture and implementation of high-availability database clusters with multi-site disaster recovery, delivering resilient infrastructure projects on time and within scope.
- **Stakeholder & Technical Alignment:** Partnered with development and operations teams as a key project stakeholder to guide technology selection and implementation, ensuring database solutions were aligned with project and business objectives.
- **Risk Management & Quality Assurance:** Strengthened project deliverables by proactively identifying risks to data integrity and security, performing root-cause analysis on issues, and implementing preventive measures to ensure high-quality outcomes.
- **Client Advisory & Solution Design:** Advised client development teams on secure and efficient database design for single-tenant and multi-tenant architectures, effectively translating business needs into technical project requirements.
- **Performance Optimization Projects:** Conducted performance tuning and optimization as a key project milestone, achieving significant improvements tailored to specific project SLAs and workload requirements.

Associate Technical Architect - Tech Mahindra Ltd., Pune, India (Dec 2005 – Apr 2014)

Key Deliverables:

- Delivered actionable business insights and improved decision-making by leveraging advanced statistical analysis and translating complex datasets into comprehensive, executive-level reports.
- Enhanced data accuracy and operational efficiency by designing and implementing scalable data collection systems across multiple business functions.
- Improved system reliability and performance by integrating data from heterogeneous sources and proactively maintaining and optimizing database environments.
- Enabled advanced analytics and large-scale reporting by developing and testing custom Hive scripts for high-volume data processing within Hadoop ecosystems.
- Reduced downtime and improved development cycles by supporting root cause analysis and troubleshooting of data processing jobs, scripts, and configurations in both standalone and pseudo-distributed environments.

EDUCATION

- **Master of Computer Applications (M.C.A.)**, Madras University, Chennai, India (2003)
- **Bachelor of Science in Computer Science**, Kakatiya University, Warangal, India (2000)